ARE100B practice final 1

Question 1

Question 1 (a): What is Ben's best response to Cameron's actions?

Ben's best response depends on Cameron's action:

If Cameron claims the armrest: Ben gets -3 if he claims, 0 if he doesn't claim. Ben's best response is to not claim.

If Cameron doesn't claim the armrest: Ben gets 1.5 if he claims, 0 if he doesn't claim. Ben's best response is to claim.

Key part of the answer here to get full marks is to make sure that Ben's best response is shown for each of Cameron's actions.

Question 1 (b): Does either player have a dominating strategy? If so, what are they? Explain your answer.

Neither player has a dominating strategy.

For Ben: His best choice depends on what Cameron does (claim when Cameron doesn't claim, don't claim when Cameron claims).

For Cameron: Similarly, her best choice depends on what Ben does (claim when Ben doesn't claim, don't claim when Ben claims).

Since each player's optimal strategy switches depending on the other player's action, neither has a strategy that is always best regardless of what the other player does.

Question 1 (c): Find all pure strategy (no randomization) Nash Equilibria. You can solve this in the payoff matrix, but make sure you write your final answer for the Nash Equilibria here.

To solve this question, it is not enough just to solve this in the payoff matrix, you must write down the Nash Equilibrium.

The answer should be the strategies/decisions each player makes in the equilibria.

Full marks would be given to:

There are two Nash equilibria. One where Ben Claims and Cameron doesn't claim; and one where Ben doesn't claim and Cameron claims.

To solve this, you might show the best response/deviations on the payoff matrices, which would look like:

Ben

		Claim armrest	Don't claim armrest
Cameron	Claim armrest	B: -3, C: -2	B: 0, C: 2
	Don't claim armrest	B: 1.5, C: 0	B: 0, C: 0

Question 1 (d): Find the mixed strategy (with randomization) Nash Equilibrium.

Solving the probabilities is not enough for the full marks - at the end, you should write down the strategies and their probabilities.

Let p_B = probability Ben claims the armrest, p_C = probability Cameron claims the armrest.

In mixed strategy equilibrium, each player must be indifferent between their pure strategies.

For Cameron to be indifferent between claiming and not claiming:

$$\begin{aligned}
1E(\pi_C|\text{Claim}) &= E(\pi_C|\text{Don't claim}) \\
p_B \times (-2) + (1-p_B) \times 2 &= p_B \times 0 + (1-p_B) \times 0 \\
-2p_B + 2 - 2p_B &= 0 \\
-4p_B &= -2 \\
p_B &= 0.5
\end{aligned}$$

For Ben to be indifferent between claiming and not claiming:

$$\begin{aligned}
E(\pi_B|\text{Claim}) &= E(\pi_B|\text{Don't claim}) \\
p_C \times (-3) + (1-p_C) \times 1.5 &= p_C \times 0 + (1-p_C) \times 0 \\
-3p_C + 1.5 - 1.5p_C &= 0 \\
-4.5p_C &= -1.5 \\
p_C &= \frac{1}{3}
\end{aligned}$$

And so, the equilibrium is:

- Ben: Claim with probability 0.5, Don't claim with probability 0.5
- Cameron: Claim with probability 1/3, Don't claim with probability 2/3

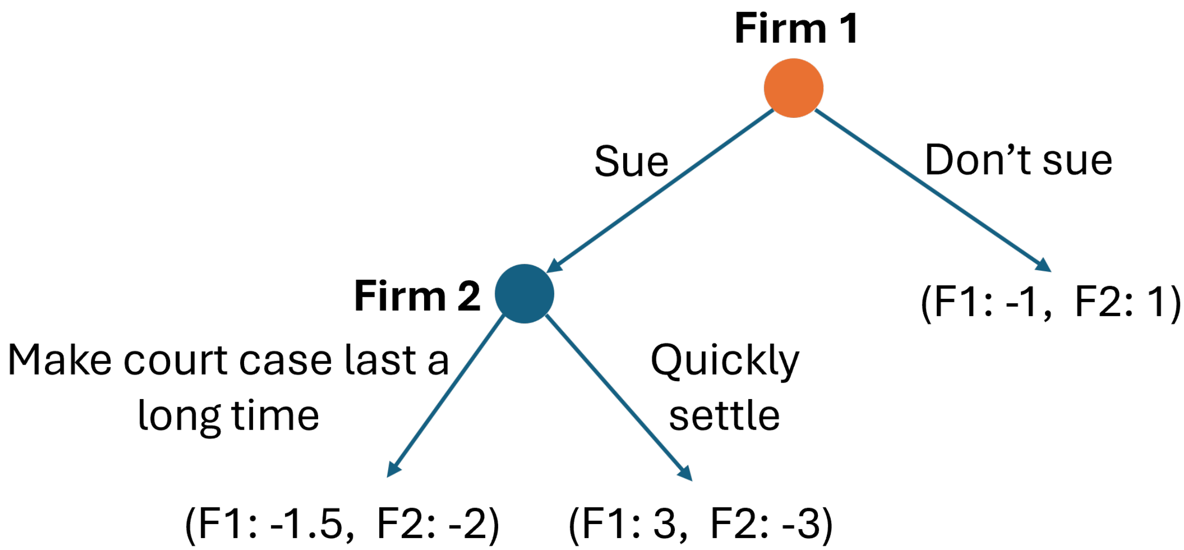
You could also write this same equilibrium as:

$$\left(\begin{pmatrix}\text{Claim} & \text{Don't claim} \\ \frac{1}{2}, & \frac{1}{2}\end{pmatrix}, \begin{pmatrix}\text{Claim} & \text{Don't claim} \\ \frac{1}{3}, & \frac{2}{3}\end{pmatrix}\right)$$

Question 2

Firm 2 is illegally entering a market where Firm 1 has regulated exclusivity. If Firm 2 enters, Firm 1 will make a loss. Firm 1 is considering suing Firm 2 for \$3. If Firm 2 gets sued, they have the option of quickly settling, and paying Firm 1 \$3; or they have the option of dragging out the court proceedings, increasing the legal costs and reducing how much Firm 1 will finally get paid. Firm 1 will make its decision first.

The following game tree shows the sequence and payoffs of this sequential/extensive form game.



Question 2 (a): Find the equilibrium outcome if Firm 1 and Firm 2 make their decisions optimally.

Solving this question just on the game tree is not enough for full marks, you should write the equilibrium outcome here.

Using backward induction to solve this sequential game:

First, analyze Firm 2's decision if Firm 1 sues:

- If Firm 2 settles quickly: Firm 2 gets -3
- If Firm 2 drags out court case: Firm 2 gets -2

Firm 2 prefers -2 over -3, so Firm 2's optimal choice is to drag out the court case.

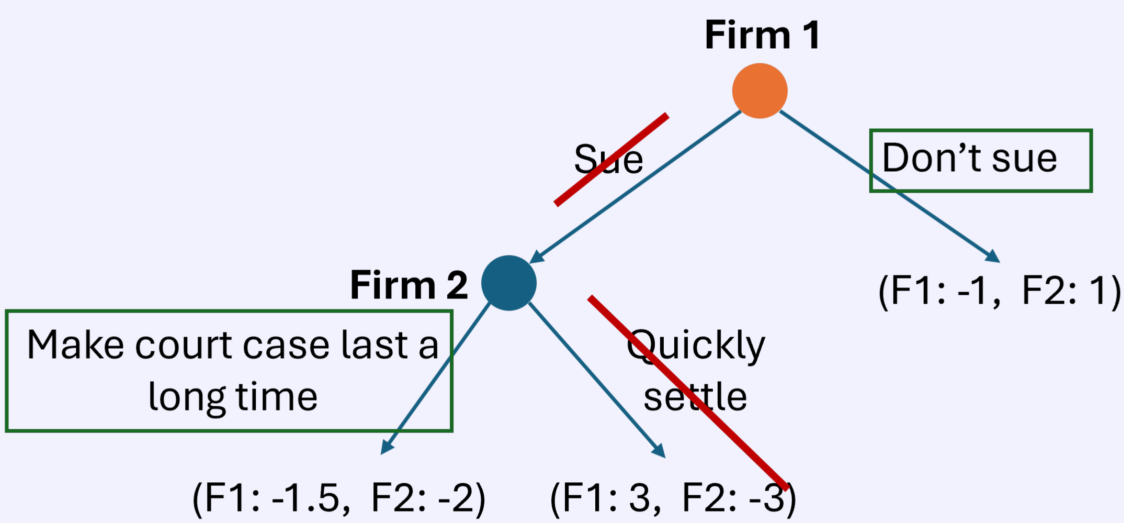
Now analyze Firm 1's initial decision, knowing Firm 2 will drag out if sued:

- If Firm 1 doesn't sue: Firm 1 gets -1
- If Firm 1 sues: Firm 2 will drag out, so Firm 1 gets -1.5

Firm 1 prefers -1 over -1.5, so Firm 1's optimal choice is to not sue.

Equilibrium outcome: Firm 1 doesn't sue, and the payoffs are (Firm 1: -1, Firm 2: 1).

The solved game tree looks like:



Question 2 (b): Write the equilibrium outcome as a subgame perfect Nash Equilibrium, that is, the optimal strategy at each decision node.

The subgame perfect Nash Equilibrium specifies the optimal action at each decision node:

Firm 1's strategy: Don't sue

Firm 2's strategy: If sued, drag out the court case

We can also write this as: (Don't sue; Drag out)

The key part of this answer is that we write out each player's optimal decision at **every** node.

Question 3

Question 3 (a): Find Sacramento Republic's profit function in terms of p_{SR} and p_A.

Sacramento Republic's demand: $q_{SR} = 12,000 - 800p_{SR} + 200p_A$ Sacramento Republic's total costs: $TC_{SR} = 5q_{SR}$

Profit = Revenue - Total Costs

$$\begin{aligned}
\pi_{SR} &= p_{SR} \times q_{SR} - TC_{SR} \\
&= p_{SR}(12,000 - 800p_{SR} + 200p_A) - 5(12,000 - 800p_{SR} + 200p_A) \\
&= 12,000p_{SR} - 800p_{SR}^2 + 200p_{SR}p_A - 60,000 + 4,000p_{SR} - 1,000p_A \\
&= 16,000p_{SR} - 800p_{SR}^2 + 200p_{SR}p_A - 1,000p_A - 60,000
\end{aligned}$$

Question 3 (b): Find the A's profit function in terms of p_{SR} and p_A.

The A's demand: $q_A = 25,000 - 600p_A + 150p_{SR}$ The A's total costs: $TC_A = 8q_A$

$$\begin{aligned}
\pi_A &= p_A \times q_A - TC_A \\
&= p_A(25,000 - 600p_A + 150p_{SR}) - 8(25,000 - 600p_A + 150p_{SR}) \\
&= 25,000p_A - 600p_A^2 + 150p_Ap_{SR} - 200,000 + 4,800p_A - 1,200p_{SR} \\
&= 29,800p_A - 600p_A^2 + 150p_Ap_{SR} - 1,200p_{SR} - 200,000
\end{aligned}$$

Question 3 (c): Find Sacramento Republic's best response function if they compete in a Bertrand game (choosing prices).

To find Sacramento Republic's best response, we maximize their profit with respect to p_{SR}:

$$\begin{aligned}
\frac{\partial \pi_{SR}}{\partial p_{SR}} &= 16,000 - 1,600p_{SR} + 200p_A = 0 \\
1,600p_{SR} &= 16,000 + 200p_A \\
p_{SR} &= 10 + 0.125p_A
\end{aligned}$$

Sacramento Republic's best response function: $p_{SR} = 10 + 0.125p_A$

Question 3 (d): Find the A's best response function if they compete in a Bertrand game (choosing prices).

To find the A's best response, we maximize their profit with respect to p_A:

$$\begin{aligned}
\frac{\partial \pi_A}{\partial p_A} &= 29,800 - 1,200p_A + 150p_{SR} = 0 \\
1,200p_A &= 29,800 + 150p_{SR} \\
p_A &= \frac{29,800 + 150p_{SR}}{1,200} \\
p_A &= 24.833 + 0.125p_{SR}
\end{aligned}$$

The A's best response function: $p_A = 24.833 + 0.125p_{SR}$

Question 3 (e): Find the Nash Equilibrium prices and quantities of tickets sold for both Sacramento Republic and the A's.

The Nash Equilibrium occurs where both teams are simultaneously playing their best responses. We solve the system of best response functions:

$$p_{SR} = 10 + 0.125p_A$$
$$p_A = 24.833 + 0.125p_{SR}$$

Substituting the second equation into the first:

$$\begin{aligned}
p_{SR} &= 10 + 0.125(24.833 + 0.125p_{SR}) \\
p_{SR} &= 10 + 3.104 + 0.0156p_{SR} \\
p_{SR} - 0.0156p_{SR} &= 13.104 \\
0.9844p_{SR} &= 13.104 \\
p_{SR} &= 13.31
\end{aligned}$$

Therefore: $p_A = 24.833 + 0.125(13.31) = 26.50$

Nash Equilibrium prices: $p_{SR} = \$13.31$, $p_A = \$26.50$

Quantities:

$$\begin{aligned}
q_{SR} &= 12,000 - 800(13.31) + 200(26.50) = 10,652 \\
q_A &= 25,000 - 600(26.50) + 150(13.31) = 11,097
\end{aligned}$$

Nash Equilibrium quantities: $q_{SR} = 10,652$ tickets, $q_A = 11,097$ tickets

Question 4

There are two banks in Davis that provide mortages to customers.

The willingnes to pay for a mortage is: $P = 8 - 0.0002Q_D$.

Bank 1 has total costs of providing mortgages of $TC_1 = 2 \times q_1$.

Bank 2 has total costs of providing mortgages of $TC_2 = 2 \times q_2$.

Question 4 (a): Find each bank's best response functions if they are competing in a Cournot game (competing by choosing quantities).

Market demand: $P = 8 - 0.0002Q_D$ where $Q_D = q_1 + q_2$ Bank 1's costs: $TC_1 = 2q_1$ (so $MC_1 = 2$) Bank 2's costs: $TC_2 = 2q_2$ (so $MC_2 = 2$)

Bank 1's profit function:

$$\begin{aligned}
\pi_1 &= P \times q_1 - TC_1 \\
&= [8 - 0.0002(q_1 + q_2)]q_1 - 2q_1 \\
&= 8q_1 - 0.0002q_1^2 - 0.0002q_1q_2 - 2q_1 \\
&= 6q_1 - 0.0002q_1^2 - 0.0002q_1q_2
\end{aligned}$$

Maximizing Bank 1's profit:

$$\begin{aligned}
\frac{\partial \pi_1}{\partial q_1} &= 6 - 0.0004q_1 - 0.0002q_2 = 0 \\
0.0004q_1 &= 6 - 0.0002q_2 \\
q_1 &= 15,000 - 0.5q_2
\end{aligned}$$

Bank 2's profit function:

$$\begin{aligned}
\pi_2 &= [8 - 0.0002(q_1 + q_2)]q_2 - 2q_2 \\
&= 6q_2 - 0.0002q_2^2 - 0.0002q_1q_2
\end{aligned}$$

Maximizing Bank 2's profit:

$$\begin{aligned}
\frac{\partial \pi_2}{\partial q_2} &= 6 - 0.0004q_2 - 0.0002q_1 = 0 \\
q_2 &= 15,000 - 0.5q_1
\end{aligned}$$

Best response functions: $q_1 = 15,000 - 0.5q_2$ and $q_2 = 15,000 - 0.5q_1$

Question 4 (b): Find the Nash equilibrium quantity of mortgages provided for each bank and the market price for mortgages in the Cournot game.

To find the Nash equilibrium, we solve the system of best response functions:

$$\begin{aligned}
q_1 &= 15,000 - 0.5q_2 \\
q_2 &= 15,000 - 0.5q_1
\end{aligned}$$

Substituting the second equation into the first:

$$\begin{aligned}
q_1 &= 15,000 - 0.5(15,000 - 0.5q_1) \\
q_1 &= 15,000 - 7,500 + 0.25q_1 \\
0.75q_1 &= 7,500 \\
q_1 &= 10,000
\end{aligned}$$

Therefore: $q_2 = 15,000 - 0.5(10,000) = 10,000$

Market price:

$$P = 8 - 0.0002(10,000 + 10,000) = 8 - 4 = 4$$

Nash equilibrium: $q_1 = q_2 = 10,000$ mortgages each, $P = \$4$

Question 4 (c): Find each bank's profit in the Nash equilibrium of the Cournot game.

Each bank's profit in equilibrium:

$$\begin{aligned}\pi_1 = \pi_2 &= P \times q_i - TC_i \\ &= 4 \times 10,000 - 2 \times 10,000 \\ &= 40,000 - 20,000 \\ &= 20,000\end{aligned}$$

Each bank's profit: $\$20,000$

Question 4 (d): The banks meet up in a dark corner at G Street Wunderbar and agree to collude to act as a cartel monopoly. They agree to split collusion output and profit 50% each. Find the collusion/cartel monopoly price, output and profit for each bank.

As a cartel, they maximize joint profits as a monopoly.

We have that the marginal cost of mortgages is: $MC = 2$. Marginal revenue is $MR = 8 - 0.0004Q$

We know that when a monopoly maximises profit we have $MC = MR$, so that monopoly quantity of mortgages is:

$$\begin{aligned}&MC = MR \\ &2 = 8 - 0.0004Q \\ &Q = 15,000\end{aligned}$$

Each bank splits this output 50% each, so $q_1 = q_2 = 7,500$

Price is then $P = 8 - 0.0002 \times 15,000 = 5$

Profit for each bank is: $\pi_1 = \pi_2 = 5 \times 7,500 - 2 \times 7,500 = 22,500$

Question 4 (e): If Bank 1 sticks to the collusive output you found in part (d), does Bank 2 have an incentive to cheat the cartel?

If Bank 1 produces the collusive quantity $q_1 = 7,500$, Bank 2's best response is:

$$q_2 = 15,000 - 0.5(7,500) = 11,250$$

Market price if Bank 2 cheats:

$$P = 8 - 0.0002(7,500 + 11,250) = 8 - 3.75 = 4.25$$

Bank 2's profit if it cheats:

$$\begin{aligned}\pi_2 &= 4.25 \times 11,250 - 2 \times 11,250 \\ &= 47,812.5 - 22,500 \\ &= 25,312.5\end{aligned}$$

Comparing profits: - Collusion: Bank 2 gets $22,500 - Cheating : Bank 2 gets$25,312.5

Since $25,312.5$ >22,500, Bank 2 has an incentive to cheat the cartel by expanding its output while Bank 1 is restricting its output!